Exponentials, Radicals, Polynomials & Rational Expressions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

If $f(x) = 5 \cdot 4^x$, what is $f(0)$?

2 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

If $3^x = 81$, what is the value of x?

3 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

What is the value of $8^{2/3}$?

4 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

If $f(x) = 12(1.5)^x$, what is the value of $f(2)$?

5 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

What is the degree of the polynomial $5x^4 - 2x^3 + 7x - 9$?

6 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

When $(x + 4)(x + 1)$ is expanded, what is the coefficient of x?

7 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Simplify $\frac{x^2 - 2x - 35}{(x + 5)}$, then evaluate the result at $x = 2$.

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

If $4^x = 64$, what is x?

9 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

When $(x + 1)(x + 6)$ is expanded, what is the coefficient of x?

10 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

A quantity $A_0 = 800$ increases by 25% to A_1 and then decreases by 20% to A_2. What is the final quantity $n = A_2$?

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11 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
When $(x-1)(x+8)$ is expanded, what is the constant term?

12 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
When $(x+4)(x+4)$ is expanded, what is the constant term?

13 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
When $(x+1)(x+7)$ is expanded, what is the constant term?

14 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
Simplify $\frac{x^2+10x+16}{(x+2)}$, then evaluate the result at $x=0$.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
If $f(x)=5\cdot 2^x$, what is $f(4)$?

16 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
What is the value of $16^{3/2}$?

17 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
When $(x+3)(x-2)$ is expanded, what is the coefficient of x?

18 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
Simplify $\frac{x^2+5x-14}{(x+7)}$, then evaluate the result at $x=4$.

19 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
When $(x-6)(x+9)$ is expanded, what is the constant term?

20 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
Simplify $\frac{x^2+11x+24}{(x+3)}$, then evaluate the result at $x=1$.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

21 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
$(2x^2+3x-1)-(x^2-2x+4)$ simplifies to

a. x^2+5x-5

b. $3x^2+x+3$

c. x^2-5x+5

d. x^2+x-5

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22 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(-8x^4)(-5x^4)$?

a. $40x^8$

b. $40x^9$

c. $40x^{16}$

d. $-13x^8$

23 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $x^9 \cdot x^2$?

a. x^{18}

b. x^{11}

c. $2x^{11}$

d. x^7

24 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Add: $(7x^2 - x + 10) + (4x^2 + 4x - 10)$

a. $13x^2 + 3x$

b. $11x^2 + 3x + 2$

c. $11x^2 + 5x$

d. $11x^2 + 3x$

25 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(b^8)^2$?

a. b^{16}

b. b^{18}

c. b^{64}

d. b^{10}

26 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(b^7)^5$?

a. b^{35}

b. b^{13}

c. b^{37}

d. b^{12}

27 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(a^6)^8$?

a. a^{14}

b. a^{50}

c. a^{48}

d. a^{15}

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28 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(y^8)^5$?

a. y^{14}

b. y^{42}

c. y^{13}

d. y^{40}

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Subtract: $(x^2 - 2x) - (4x^2 - 8x - 9)$

a. $-3x^2 + 6x + 11$

b. $-3x^2 + 8x + 9$

c. $-x^2 + 6x + 9$

d. $-3x^2 + 6x + 9$

30 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Subtract: $(5x^2 - 4x + 3) - (4x^2 + 4x - 3)$

a. $x^2 - 8x + 6$

b. $x^2 - 8x + 8$

c. $3x^2 - 8x + 6$

d. $x^2 - 6x + 6$

31 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(-6x^1)(8x^3)$?

a. $-48x^5$

b. $2x^4$

c. $-48x^3$

d. $-48x^4$

32 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Add: $(7x^2 - 7x - 2) + (2x^2 + 3x - 6)$

a. $9x^2 - 2x - 8$

b. $9x^2 - 4x - 8$

c. $9x^2 - 4x - 6$

d. $11x^2 - 4x - 8$

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33 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Subtract: $(6x^2 + 5x - 3) - (2x^2 + 2x - 5)$

a. $6x^2 + 3x + 2$

b. $4x^2 + 3x + 4$

c. $4x^2 + 5x + 2$

d. $4x^2 + 3x + 2$

34 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(x^6)^9$?

a. x^{16}

b. x^{56}

c. x^{54}

d. x^{15}

35 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $b^7 \cdot b^4$?

a. b^{28}

b. b^3

c. $2b^{11}$

d. b^{11}

36 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Subtract: $(5x^2 - 8x - 7) - (6x^2 - 3x + 9)$

a. $-x^2 - 5x - 16$

b. $-x^2 - 5x - 14$

c. $x^2 - 5x - 16$

d. $-x^2 - 3x - 16$

37 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $x^{1/3}$ for $x \geq 0$?

a. $3x$

b. $\frac{x}{3}$

c. $\sqrt[3]{x}$

d. $\frac{1}{x^3}$

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(-4x^4)(-6x^2)$?

a. $24x^7$

b. $-10x^6$

c. $24x^6$

d. $24x^8$

39 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $x^4 \cdot x^7$?

a. x^3

b. x^{28}

c. x^{11}

d. $2x^{11}$

40 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Add: $(8x^2 - 4x + 3) + (2x^2 - 7x - 8)$

a. $10x^2 - 9x - 5$

b. $12x^2 - 11x - 5$

c. $10x^2 - 11x - 5$

d. $10x^2 - 11x - 3$

41 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Subtract: $(2x^2 + 4x + 7) - (6x^2 + x + 3)$

a. $-4x^2 + 3x + 4$

b. $-2x^2 + 3x + 4$

c. $-4x^2 + 5x + 4$

d. $-4x^2 + 3x + 6$

42 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(a^5)^3$?

a. a^{15}

b. a^8

c. a^9

d. a^{17}

43 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Add: $(8x^2 - 3x + 5) + (2x^2 - x - 3)$

a. $12x^2 - 4x + 2$

b. $10x^2 - 4x + 4$

c. $10x^2 - 2x + 2$

d. $10x^2 - 4x + 2$

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44 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Add: $(7x^2 - 7x + 4) + (5x^2 + x + 4)$

a. $12x^2 - 6x + 10$

b. $14x^2 - 6x + 8$

c. $12x^2 - 4x + 8$

d. $12x^2 - 6x + 8$

45 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which of the following is equivalent to $(x^3)^4$?

a. x^7

b. x^{12}

c. x^{34}

d. x^{81}

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